

Sparse Word Representation

Dr. Saeedeh Momtazi



دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)



آکادمی‌ها



➤ **Word Concept**

➤ **Word Vectors**



What do words mean?

➤ First thought: look in a dictionary



Words, Lemmas, Senses, Definitions

lemma

pepper, n.

Pronunciation: Brit fix pal fimpar Forms: OE peopor (rare), OE pipcer (transrission error), OE pipor, OE pipur (rare Frequency (in current use): Etymology AborrawingfromLatin Etmon: Lath piper < dasical Latin piper, a loatmord < Ink-Aran (w is anzient Greek yrerzept); conpare Sa3

1. The spice or the plant i.

a. A hot pungent spice derived from the prepared fruits (peppercorns) of the pepper plant, *Piper nigrunz* (see sense 2a), used from early times to season food, either whole or ground to powder (often in association with salt). Also (locally, chiefly with distinguishing word): a similar spice derived from the fruits of certain other spesdes of the genus *Piper*; the fruits themselves. The ertund spice :min *Piper nignen* come. in ma 5arms, the more purgent Ma& pepper, produced from bhdtpfepperconn, and the raider Lute pepper, produced from whieppepercann: see MAC.. adj. and a special use. sa, nen 'Aix. n a, and tun aci and it 'special use. t(a).

2.

a. The plant *Piper nigrum* (family *Piperaceae*), a climbing shrub indigenous to South Asia and also cultivated elsewhere in the tropics, which has alternate stalked entire leases, with pendulous spikes of small green flowers opposite the leaves, succeeded by small berries turning red when ripe. Also more widely: any plant of the genus *Piper* or the family *P iperaceae*.

b. Usu. with distinguishing word: any of numerous plant of other families having hot pungent fruits or leases which resemble pepper (La) in taste and in some cases are used as a substitute for it

c. U.S. The California pepper tree, *Schinus molle*. Cf. PEPPER TREE n. 3.

definition

3. Any of various forms of capsicum, esp. *Capsicum annum* var. *annuum*. Originally (chiefly with distinguishing word): any variety of the C. annum Longum group, with elongated fruits having a hot, pungent taste, the source of cayenne, chilli powder, paprika, etc., or of the perennial C. frutescens, the source of Tabasco sauce. Nov frequently (more fully sweetpeppen: any variety of the C. annum Grossum group, with large, bell-shaped or apple-shaped, mild-flavoured fruits, usually ripening to red, orange, or yellow and eaten raw in salads or cooked as a vegetable. Also the fruit of any of these capsicums. Sineetpewers are °Rai:used in theirkreen immature state (rmre fullygreenpepprr), bit some new variet remakt green when ripe.

sense



Lemma "pepper"

- Sense 1: spice from pepper plant
- Sense 2: the pepper plant itself
- Sense 3: another similar plant (Jamaican pepper)
- Sense 4: another plant with peppercorns (California pepper)
- Sense 5: capsicum (i.e. chili, paprika, bell pepper, etc)



- A sense or "concept" is the meaning component of a word
- There are relations between senses



Relation: Synonymy

➤ Synonyms have the same meaning in some or all contexts.

- ▶ filbert / hazelnut
- ▶ couch / sofa
- ▶ big / large
- ▶ automobile / car
- ▶ vomit / throw up
- ▶ water / H₂O



Relation: Synonymy

- Note that there are probably no examples of perfect synonymy.
 - ▶ Even if many aspects of meaning are identical
 - ▶ Still may not preserve the acceptability based on notions of politeness, slang, register, genre, etc.

- The Linguistic Principle of Contrast:
 - ▶ Difference in form -> difference in meaning



Relation: Synonymy?

- Water/H₂O
- Big/large
- Brave/courageous



Relation: Antonym

➤ Senses that are opposites with respect to one feature of meaning

➤ Otherwise, they are very similar!

- ▶ hot/cold
- ▶ dark/light
- ▶ short/long
- ▶ fast/slow
- ▶ rise/fall
- ▶ up/down
- ▶ in/out

➤ More formally: antonyms can

- ▶ **define a binary opposition or be at opposite ends of a scale**
- ▶ long/short, fast/slow
- ▶ **Be reversives:** rise/fall, up/down



Relation: Similarity

- Words with similar meanings.
- Not synonyms, but sharing some element of meaning

▶ car	▶ bicycle
▶ cow	▶ horse

► Ask humans how similar 2 words are

word1	word2	similarity
vanish	disappear	9.8
behave	obey	7.3
belief	impression	5.95
muscle	bone	3.65
modest	flexible	0.98
modest	agreement	0.3

► SimLex-999 dataset (Hill et al., 2015)



Relation: Word Relatedness

- Also called "word association"
- Words be related in any way, perhaps via a semantic frame or field
 - ▶ car, bicycle: **similar**
 - ▶ car, gasoline: **related**, not similar



» Words that

- ▶ cover a particular semantic domain
- ▶ bear structured relations with each other.

» hospitals

- ▶ surgeon, scalpel, nurse, anesthetic, hospital

» restaurants

- ▶ waiter, menu, plate, food, menu, chef

» houses

- ▶ door, roof, kitchen, family, bed



Relation: Superordinate/ Subordinate

➤ One sense is a **subordinate** of another if the first sense is more specific, denoting a subclass of the other

- ▶ car is a subordinate of vehicle
- ▶ mango is a subordinate of fruit

➤ Conversely **superordinate**

- ▶ vehicle is a superordinate of car
- ▶ fruit is a superordinate of mango

Superordinate	Vehicle	Fruit	Furniture
Subordinate	Car	Mango	Chair



Category Levels

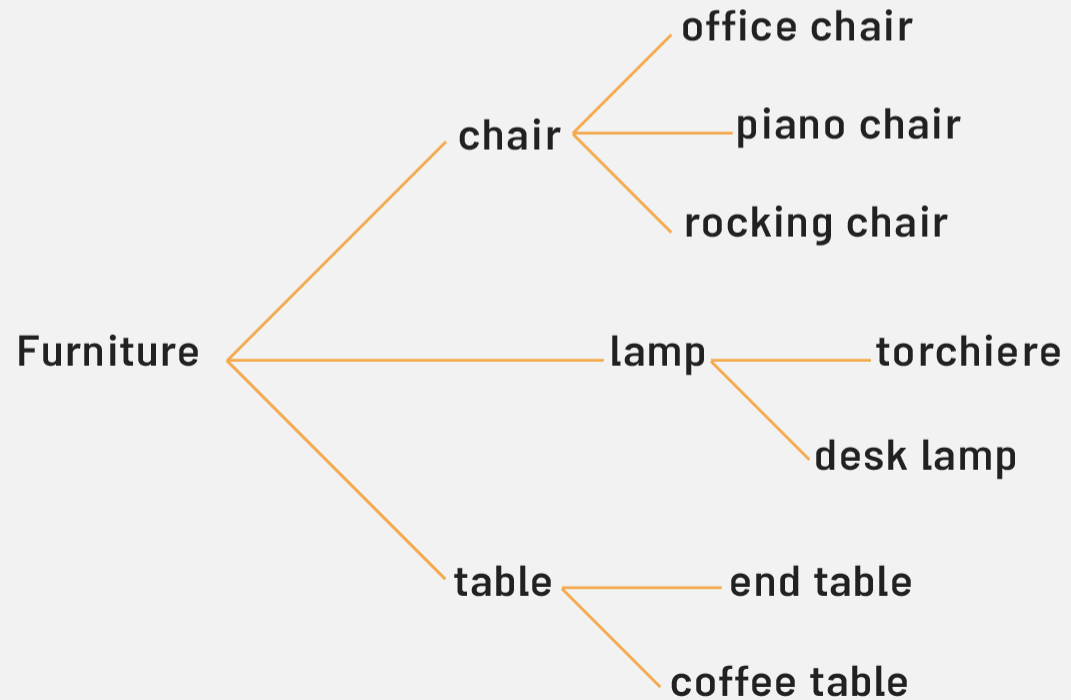
- These levels are not symmetric
- One level of category is distinguished from the others
- Name these items





Category Levels

Superordinate Basic Subordinate





Cluster of Interactional Properties

- Basic level things are "human-sized"
- Is the level which is learned earliest and at which things are first named
- It is the level at which names are shortest and used most frequently

- Consider chairs
 - ▶ We know how to interact with a chair (sitting)
 - ▶ Not so clear for superordinate categories like furniture
 - ▶ "Imagine a furniture without thinking of a bed/table/chair/specific basic-level category"



➤ Concepts or word senses

- ▶ Have a complex many-to-many association with words (homonymy, multiple senses)

➤ Have relations with each other

- | | |
|--------------|-----------------------------|
| ▶ Synonymy | ▶ Relatedness |
| ▶ Antonymy | ▶ Superordinate/subordinate |
| ▶ Similarity | ▶ Connotation |

➤ But how to define a concept?



Classical ("Aristotelian") Theory of Concepts

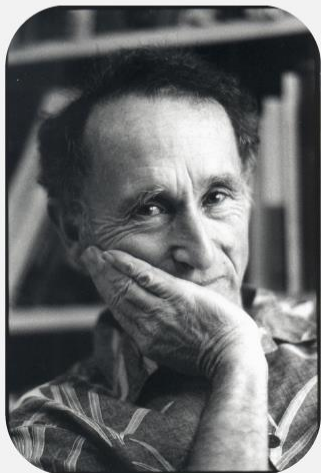
- The meaning of a word
 - ▶ a concept defined by necessary and sufficient conditions
- A necessary condition for being an X is a condition C that X must satisfy in order for it to be an X.
 - ▶ If not C, then not X
 - ▶ "Having four sides" is necessary to be a square.
- A sufficient condition for being an X is condition such that if something satisfies condition C, then it must be an X.
 - ▶ If and only if C, then X



Complex and Context-dependent Features

➤ Problem 1: The features are complex and may be context-dependent

➤ What are these? Cup or bowl?



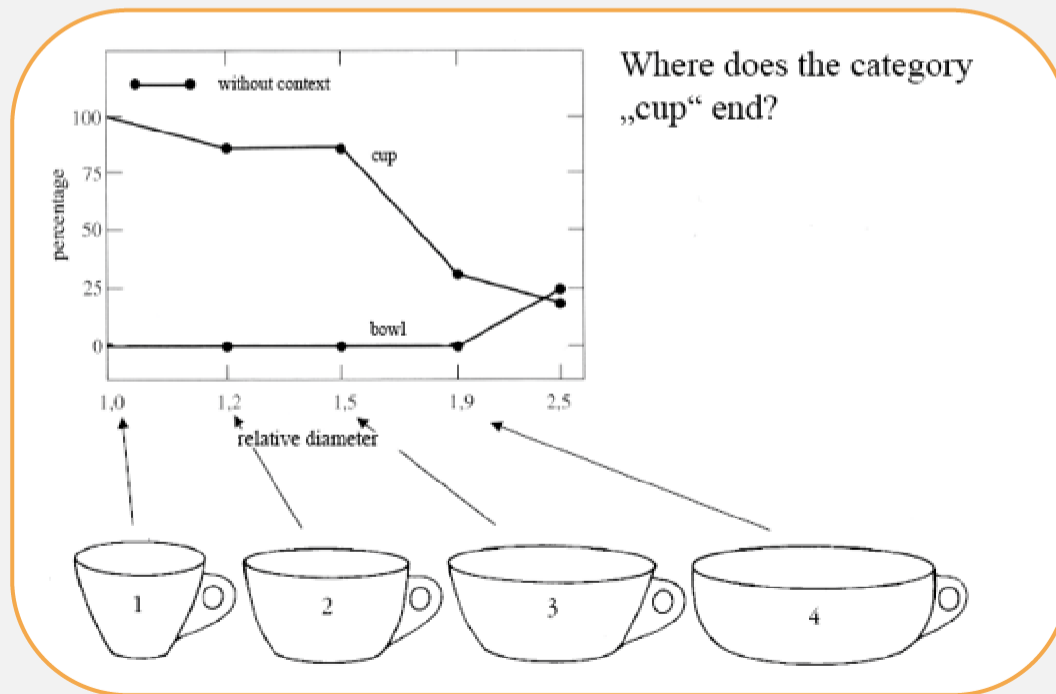
William Labov. 1975





Feature-based Definition

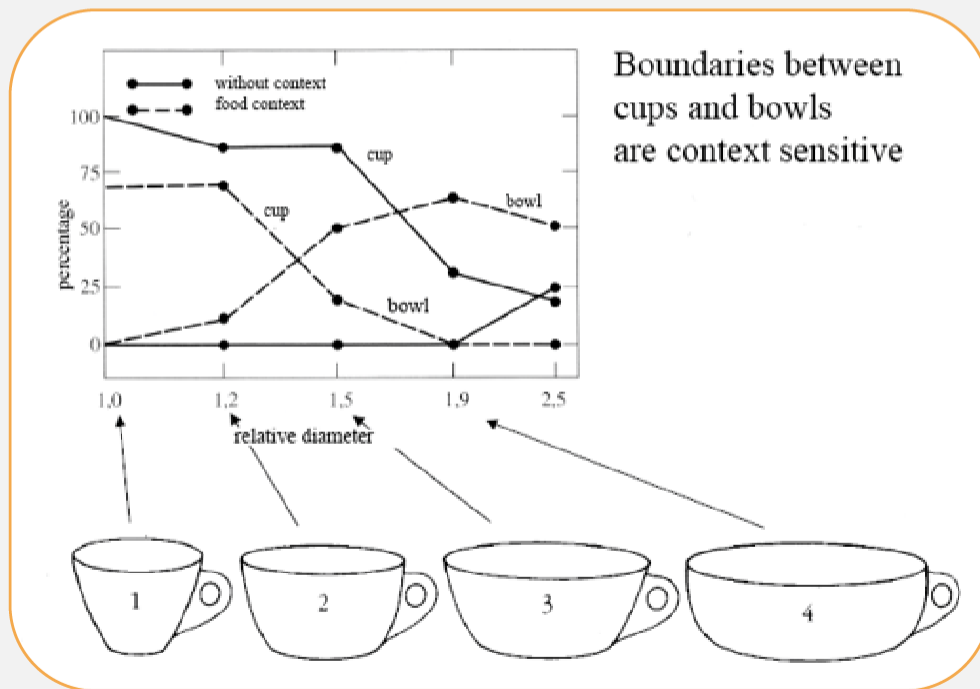
➤ The category depends on complex features of the object (diameter, etc)





Context-based Definition

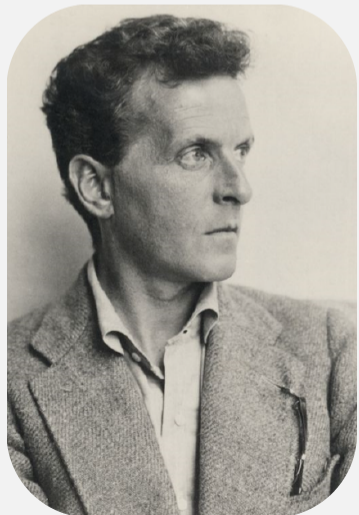
➤ The category depends on the context! (If there is food in it, it's a bowl)





Ludwig Wittgenstein (1889-1951)

- Philosopher of language
- In his late years, a proponent of studying "ordinary language"
- "The meaning of a word is its use in the language"





Defining Words by their Usages

- In particular, words are defined by their environments (the words around them)
- Zellig Harris (1954): If A and B have almost identical environments we say that they are synonyms.



What does "ABC" Means?

- A plate full of ABC is on the table
- Everybody likes ABC
- ABC is cooked within 30 minutes
- We make ABC with potato



What does "ABC" Means?

- A plate full of ABC is on the table
- Everybody likes ABC
- ABC is cooked within 30 minutes
- We make ABC with potato

→ A kind of food





- We'll build a new model of meaning focusing on similarity
- Each word = a vector
- Similar words are "nearby in space"

to a you
with
by that is
i now
than are

dislike
not good bad
worst
incredibly bad worse

terrific amazing
very good good nice
incredibly good fantastic wonderful



- We define a word as a vector
- Called an "embedding" because it's embedded into a space
- The standard way to represent meaning in NLP

- Fine-grained model of meaning for similarity
 - ▶ NLP tasks like sentiment analysis
 - ▶ With words, requires same word to be in training and test
 - ▶ With embeddings: ok if similar words occurred!!!
 - ▶ Question answering, conversational agents, etc



➤ Word Concept

➤ **Word Vectors**



➤ We'll introduce 2 kinds of embeddings

➤ Sparse vector representation

- ▶ A common baseline model
- ▶ Words are represented by a simple function of the counts of nearby words
- ▶ Tf-idf

➤ Dense vector representation

- ▶ Representation is created by training a model to distinguish nearby and far-away words
- ▶ Word2vec and Glove



- Words
- Vectors
- Co-occurrence matrices



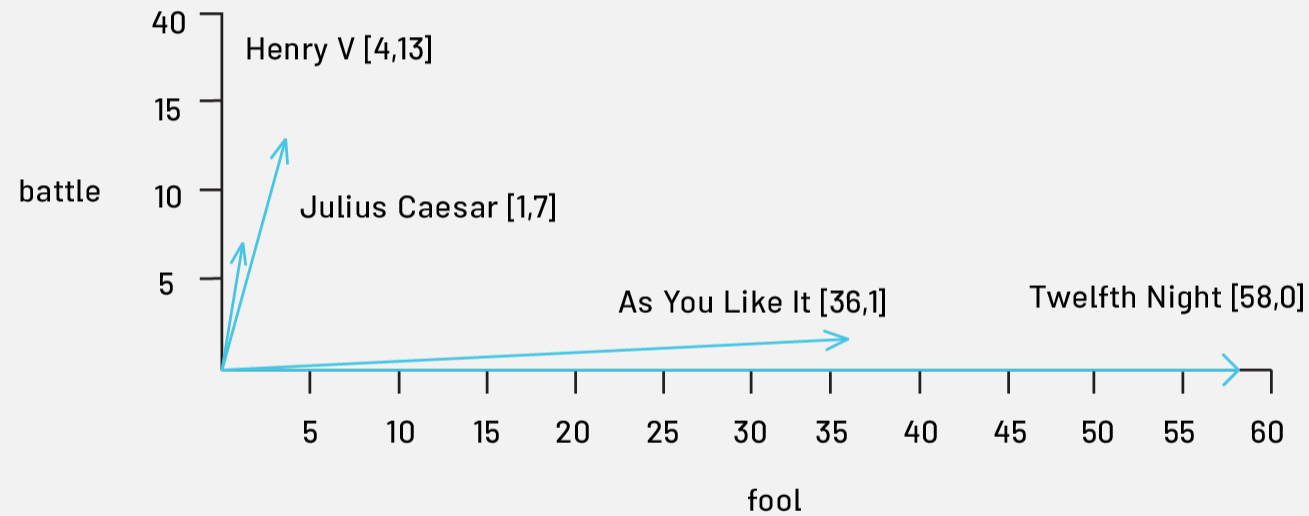
Term-document Matrix

➤ Each document is represented by a vector of words

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3



Visualizing Document Vectors





Vectors are the Basis of Information Retrieval

- Vectors are similar for the two comedies
- Different than the history
- Comedies have more fools and wit and fewer battles.

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3



Words can be Vectors too

- ▶ battle is "the kind of word that occurs in Julius Caesar and Henry V"
- ▶ fool is "the kind of word that occurs in comedies, especially Twelfth Night"

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3



More Common: Word-Word Matrix (or "Term-Context Matrix")

➤ Two words are similar in meaning if their context vectors are similar

sugar, a sliced lemon, a tablespoonful of
their enjoyment. Cautiously she sampled her first
well suited to programming on the digital
for the purpose of gathering data and

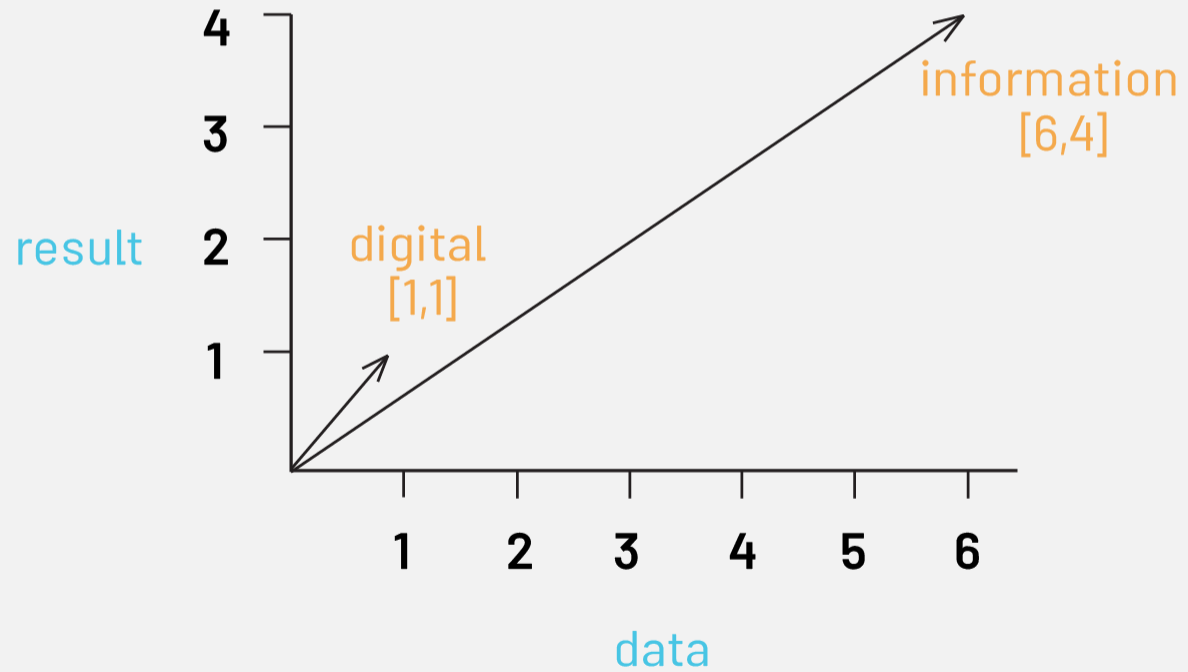
apricot
pineapple
computer.
information

jam, a pinch each of,
and another fruit whose taste she likened
In finding the optimal R-stage policy from
necessary for the study authorized in the

	aardvark	computer	data	pinch	result	sugar
apricot	0	0	0	1	0	1
pineapple	0	0	0	1	0	1
digital	0	2	1	0	1	0
information	0	1	6	0	4	0



More Common: Word-Word Matrix (or "Term-Context Matrix")





Cosine for Computing Similarity

- -1: vectors point in opposite directions
 - +1: vectors point in same directions
 - 0: vectors are orthogonal
-
- Frequency is non-negative, so cosine range 0-1

$$\text{cosine}(\vec{v}, \vec{w}) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$



Cosine for Computing Similarity

➤ Example

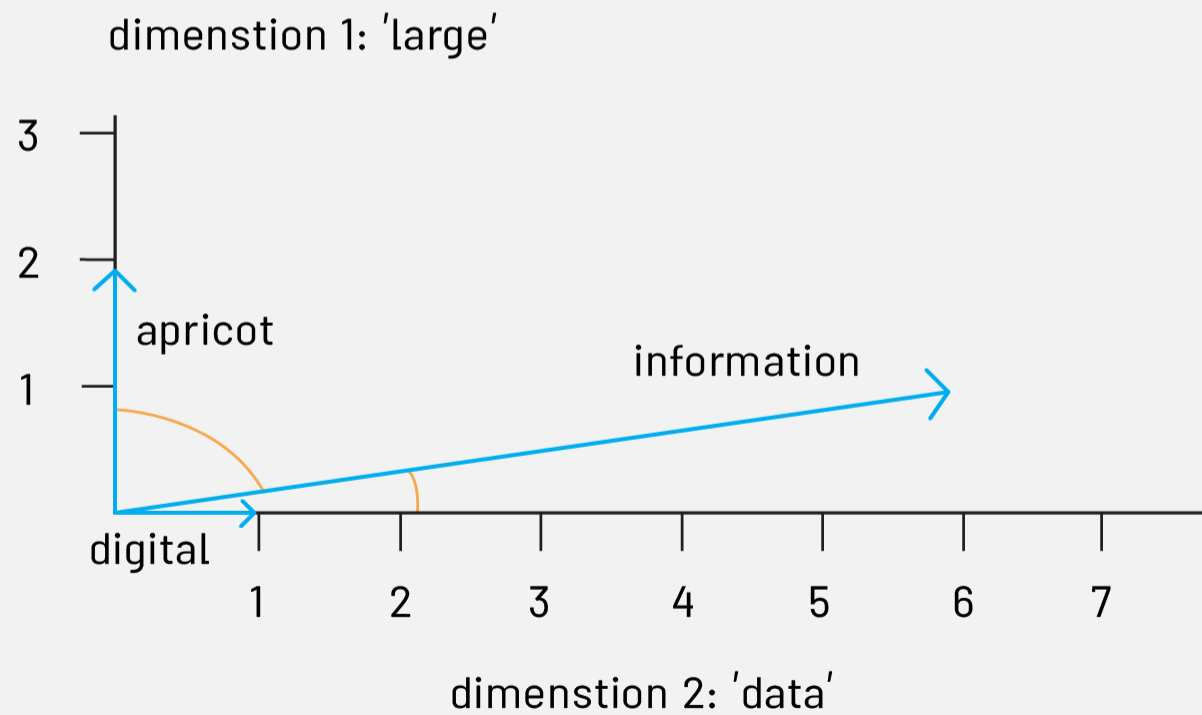
➤ Which pair of words is more similar?

$$\begin{aligned} \text{▶ cosine(apricot, information)} &= \frac{1+0+0}{\sqrt{1+0+0} \sqrt{1+36+1}} = \frac{1}{\sqrt{38}} = .16 \\ \text{▶ cosine(digital, information)} &= \frac{0+6+2}{\sqrt{1+36+1} \sqrt{1+4+1}} = \frac{8}{\sqrt{38} \sqrt{5}} = .58 \\ \text{▶ cosine(apricot, digital)} &= \frac{0+0+0}{\sqrt{1+0+0} \sqrt{0+1+4}} = 0 \end{aligned}$$

	Large	Data	Computer
apricot	1	0	0
digital	0	1	2
information	1	6	1



Visualizing Cosines





Raw Frequency

- But raw frequency is a bad representation
- Frequency is clearly useful; if sugar appears a lot near apricot, that's useful information.
- But overly frequent words like the, it, or they are not very informative about the context.
- Need a function that resolves this frequency paradox!



tf-idf: Combine Two Factors

➤ **tf:** term frequency. frequency count (usually log-transformed)

$$\text{tf}_{t,d} = \begin{cases} 1 + \log_{10} \text{count}(t, d) & \text{if } \text{count}(t, d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

➤ **idf:** inverse document frequency

$$\text{idf}_i = \log \left(\frac{N}{\text{df}_i} \right)$$

Diagram illustrating the components of the idf formula:

- An arrow points from the text "Total # of docs in collection" to the variable N in the numerator.
- An arrow points from the text "# of docs that have word i" to the variable df_i in the denominator.

➤ **tf-idf value for word t in document d**

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$



Summary: tf-idf

- Compare two words using tf-idf cosine to see if they are similar
- Compare two documents
 - ▶ Take the centroid of vectors of all the words in the document
 - ▶ Centroid document vector is:

$$d = \frac{w_1 + w_2 + \dots + w_k}{k}$$



Positive Pointwise Mutual Information (PPMI)

- An alternative to tf-idf
- Ask whether a context word is **particularly informative** about the target word.



Pointwise Mutual Information

➤ Pointwise mutual information

- ▶ Do events x and y co-occur more than if they were independent?

$$\text{PMI}(X, Y) = \log_2 \frac{P(x, y)}{P(x)P(y)}$$

➤ PMI between two words

- ▶ Do words x and y co-occur more than if they were independent?

$$\text{PMI}(\text{word}_1, \text{word}_2) = \log_2 \frac{P(\text{word}_1, \text{word}_2)}{P(\text{word}_1)P(\text{word}_2)}$$



Positive Pointwise Mutual Information

- PMI ranges from $-\infty$ "to" $+\infty$
- But the negative values are problematic
 - ▶ Things are co-occurring less than we expect by chance
 - ▶ Unreliable without enormous corpora
 - ▶ Imagine w_1 and w_2 whose probability is each 10^{-6}
 - ▶ Hard to be sure $p(w_1, w_2)$ is significantly different than 10^{-12}
 - ▶ Plus it's not clear people are good at "unrelatedness"
- So we just replace negative PMI values by 0
- Positive PMI (PPMI) between word1 and word2

$$\text{PPMI}(\text{word}_1, \text{word}_2) = \max\left(\log_2 \frac{P(\text{word}_1, \text{word}_2)}{P(\text{word}_1)P(\text{word}_2)}, 0\right)$$



Count (w,context)

	computer	data	pinch	result	sugar
apricot	0	0	1	0	1
pineapple	0	0	1	0	1
digital	2	1	0	1	0
information	1	6	0	4	0

PPMI (w,context)

	computer	data	pinch	result	sugar
apricot	-	-	2.27	0	2.25
pineapple	-	-	2.27	0	2.25
digital	1.66	0.00	-	0.00	-
information	0.00	0.57	-	0.47	-

P (w,context)

	computer	data	pinch	result	sugar
apricot	0.00	0.00	0.05	0.00	0.05
pineapple	0.00	0.00	0.05	0.00	0.05
digital	0.11	0.05	0.00	0.05	0.00
information	0.05	0.32	0.00	0.21	0.00

P (w)

0.11

0.11

0.21

0.58

P (context)

0.16

0.37

0.11

0.26

0.11



Weighting PMI

➤ PMI is biased toward infrequent events

- ▶ Very rare words have very high PMI values

➤ Two solutions

- ▶ Give rare words slightly higher probabilities
- ▶ Use add-one smoothing (which has a similar effect)



- Survey of Lexical Semantics
- Idea of embeddings: Represent a word as a function of its distribution with other words
 - ▶ Tf-idf
 - ▶ PPMI
- Next lecture: sparse vs dense embeddings, word2vec



Further Reading

➤ Speech and Language Processing (3rd ed. draft)

▶ Chapter 6